

Problem 8.17. Let A be the language of properly nested parentheses. For example, $(())$ and $((()()))()$ are in A , but $)()$ is not. Show that A is in L .

Proof. To show that $A \in L$, we give a log space algorithm M that decides A .

$M =$ “On input $\langle w \rangle$, where w is a string:

1. Let c be a binary integer initialized with 0.
2. Repeat for each symbol $b = w_1w_2 \dots w_n$.
3. If b is $($, then increment c by 1.
4. If b is $)$ and $c > 0$, then increment c by 1.
5. If b is $)$ and $c = 0$, then *reject*.
6. If $c = 0$, then *accept*. Otherwise, *reject*.”

□